

The tissue removed from this sunflower plant will be used in genetic research at the Sungene Technologies Laboratory in Palo Alto, California.



ecology. SEE ALSO AGRICULTURAL BIOTECHNOLOGY; *ARABIDOPSIS THALIANA*; BIOPESTICIDES; BIOTECHNOLOGY ENTREPRENEUR; COLLEGE PROFESSOR; GENETICALLY MODIFIED FOODS; GENETICIST; LABORATORY TECHNICIAN; MAIZE; MOLECULAR BIOLOGIST.

Ricki Lewis

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Plasmid

Plasmids are naturally occurring, stable genetic elements found in bacteria, fungi, and even in the mitochondria of some plants. They may be composed of DNA or RNA, double-stranded or single-stranded, linear or circular.

Plasmids almost always exist and replicate independently of the chromosome of the cell in which they are found.

Types of Plasmids

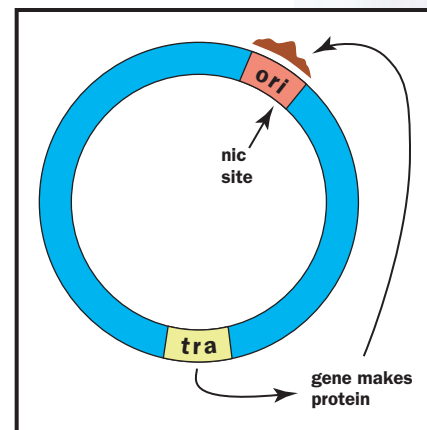
Plasmids are not usually required by their host cell for its survival. Instead, they carry genes that confer a selective advantage on their host, such as resistance to heavy metals or resistance to naturally made antibiotics carried by other organisms. Alternatively, they may produce antibiotics (toxins) that help the host to compete for food or space. For instance, antibiotic resistance genes produced by a plasmid will allow its host bacteria to grow even in the presence of competing bacteria or fungi that produce these antibiotics.

Plasmids are subgrouped into five main types based on phenotypic function. R plasmids carry genes encoding resistance to antibiotics. Col plasmids confer on their host the ability to produce antibacterial polypeptides called bacteriocins that are often lethal to closely related or other bacteria. The col proteins of *E. coli* are encoded by plasmids such as ColE1. F plasmids contain the F or fertility system required for conjugation (the transfer of genetic information between two cells). These are also known as episomes because, under some circumstances, they can integrate into the host chromosome and thereby promote the transfer of chromosomal DNA between bacterial cells. Degradative or catabolic plasmids allow a host bacterium to metabolize normally undegradable or difficult compounds such as various pesticides. Finally, virulence plasmids confer **pathogenicity** on a host organism by the production of toxins or other virulence factors.

Replication

One common feature of all plasmids is a specific sequence of nucleotides termed an origin of replication (*ori*). This sequence, together with other regulatory sequences, is referred to as a replicon. The replicon allows a plasmid to replicate within a host cell independently of the host cell's own replication cycle. If the plasmid makes many copies of itself per cell, it is termed a "relaxed" plasmid. If it maintains itself in fewer numbers within the cell it is termed a "stringent" plasmid. Two different plasmids can coexist in the same cell only if they share the same replication elements. If they do not, they will be unable to be propagated stably in the same cell line, and are termed incompatible.

In nature, plasmid inheritance can occur through a variety of mechanisms. During conjugation between two bacterial strains, plasmids can be transferred along with the bacterial DNA, and this activity is controlled by a set of transfer (*tra*) genes that are located on the plasmid and not on the bacterial chromosome. The proteins produced by these transfer genes bind to the DNA at the *ori* site to form a DNA-protein complex known as a relaxosome. This complex makes a nick, or break, in one of the two strands of the double-stranded plasmid DNA molecule. The place where this break occurs is called the "nic" site, and the nicked DNA is said to be "relaxed" because the DNA unwinds as a result of the nick in one of the strands. The single-stranded DNA that is generated by the nick is thought to be unwound and transferred through the pilus, or mating bridge, that connects the two bacteria entering the recipient bacteria. The other strand is left in the donor



The transfer gene (*tra*) makes a protein that binds to the origin of replication site (*ori*). The protein nicks the DNA, relaxing it and allowing it to be transferred to another cell.

pathogenicity ability to cause disease

bacteria. It acts as a template for the synthesis of a new complementary DNA strand forming a double-stranded plasmid DNA molecule.

Some nonconjugative plasmids can also be transferred into bacteria by means of a process called mobilization, as long as they carry the necessary (*mob*) genes. Others are taken up by bacterial cells during the process known as transformation. Finally, plasmids that exist in a host cell that undergoes fission (cell division) are simply divided between the resultant two daughter cells.

Use in Research and Technology

Because of their ability to move genes from cell to cell, plasmids have become versatile tools for both research and biotechnology. In the laboratory, researchers use plasmids to carry marker genes, allowing them to trace the plasmid's inheritance across host cells. Transferred or "cloned" genes are used to produce a variety of important medical, agricultural, or environmental products that can be economically used by humans.

Researchers have also engineered plasmids to be extremely efficient cloning **vectors**. To be used in this way, the plasmid must contain at least one origin of replication, a multiple cloning site (called a polylinker) where a variety of **restriction enzymes** can cut so that foreign DNA can be inserted, a selectable genetic marker, and **transcription** and **translation** signals recognized by the host cell, so that the expression of a cloned gene can be easily identified.

The foreign DNA is often inserted in such a way that the expression of the foreign gene is tied to the expression of a marker gene. For example, one of the most popular methods to show that a foreign DNA has been inserted and expressed in the host is by the insertional inactivation of the *lac Z* gene. In this case, the foreign DNA is inserted in the middle of the *lac Z* gene so that the gene becomes defective and the enzyme it codes for no longer works. The damaged enzyme therefore cannot **cleave** the artificial substrate Xgal to produce a blue color or blue colony, as it normally would, and white colonies of bacteria are produced. Therefore, the white colonies indicate that artificial DNA has been successfully cloned or recombined into the plasmid in the *lac Z* gene, whereas nonrecombinant colonies are blue. The white colonies can thus be easily isolated for further expansion and experimentation.

Under certain circumstances, recombinant DNA experiments using plasmids are considered to be hazardous, and the ease with which plasmids are acquired by bacteria has led them to be classed as biohazards. They are therefore subject to guidelines and may require registration and approval. A publication produced by the National Institutes of Health, titled *Guidelines for Research Involving Recombinant DNA Molecules*, is the definitive reference for recombinant DNA research in the United States and should be consulted when considering research, particularly biomedical research, involving plasmids. SEE ALSO ANTIBIOTIC RESISTANCE; CLONING GENES; CONJUGATION; INHERITANCE, EXTRANUCLEAR; MARKER SYSTEMS; RESTRICTION ENZYMES; TRANSFORMATION.

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vectors carriers

restriction enzymes
enzymes that cut DNA
at a particular sequence

transcription messen-
ger RNA formation from
a DNA sequence

translation synthesis of
protein using mRNA
code

cleave split

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Pleiotropy

Pleiotropy is the phenomenon whereby a single gene has multiple consequences in numerous tissues. Pleiotropic effects stem from both normal and mutated genes, but those caused by mutations are often more noticeable and easier to study. Pleiotropy is actually more common than its opposite, since in a complex organism, a protein from a single gene is likely to be expressed in more than one tissue, and the cascade of problems caused by a mutation is likely to lead to numerous complications throughout the organism. Single-gene defects with effects in only one tissue are more common for nonessential features such as hair texture or eye color.

Sickle cell disease is a classic example of pleiotropy. This disease develops in persons carrying two defective alleles for a blood protein, beta-hemoglobin. Mutant beta-hemoglobins are misaligned inside a blood cell and cause misshapen red blood cells at low oxygen concentrations. Deformed blood cells impair circulation. Impaired circulation damages kidneys and bone. In this case, the gene defect itself only affects one tissue, the blood. The consequences of that defect are found in other tissues and organs.

One baby in three thousand to four thousand births is born with neurofibromatosis, an **autosomal dominant** disease caused by mutation in a tumor suppressor gene that helps regulate cell division and cell-cell contacts. A truncated version of the tumor suppression protein, neurofibromin (NF I) is implicated in the disease. This mutant protein can come from missense or nonsense mutations, or from reading-frame shifts after a repetitive element called Alu is inserted upstream of the NF I reading frame (a reading frame is the DNA that codes for proteins). Because the mutant protein is unable to regulate cell division, tumors grow on the nerves throughout the body. The tumors produce collateral damage: low blood sugar, intestinal bleeding, café-au-lait spots on the skin, mental retardation, heart problems, high blood pressure, fractures, spinal cord lesions, blindness, aneurysms, arthritis, and respiratory distress.

autosomal dominant
pattern of inheritance in which inheritance of a single allele from either parent results in expression of the trait

Signaling Pathways

Many pleiotropic conditions arise from genes whose products are involved in signaling and regulation pathways. Because these proteins coordinate daily life in numerous tissues, defects in them have numerous consequences, as one breakdown leads to another.

Myotonic dystrophy is another autosomal dominant disorder. A gene for a protein—a **kinase**, involved in signalling and communication within the cell—is burdened with up to three thousand extra pieces of DNA. The extra DNA comes from trinucleotides (CTG)_n that are added by mistake during DNA duplication, both in germ line cells and during early cell divi-

kinase an enzyme that adds a phosphate group to another molecule, usually a protein



The café-au-lait spots typical of neurofibromatosis are evident on this patient's hand.

sions in the embryo. During transcription and translation, the kinase is not put together right, and the kinase's work in muscles goes badly. Muscles contract but cannot relax quickly. Young persons may have heart attacks, generalized muscle weakness, and loss of bulk. Swallowing and speech is hard, due to weak muscles in the tongue and neck. Other pleiotropic effects include baldness, cataracts, and changes in intelligence.

Pleiotropic outcomes are common with hormones. Hormones are signals that create multiple responses in tissues that carry receptors for them. The receptor binds to the hormone and triggers a cascade of reactions inside the cell. A defective receptor loses or misinterprets the signal. When the hormone insulin meets defective insulin receptors on an individual's cells, the person is more likely to develop type II diabetes. Cells do not open their gateways to let sugar in from the bloodstream, and the cells almost starve to death in the midst of plenty. Meanwhile, sugar accumulates in the blood and causes all sorts of ramifications for blood circulation, and it damages capillaries in all areas, from kidneys, to eyes, to feet. Gangrene, mental disturbances, kidney failure, and blindness can and do occur. Diminished give-and-take of sugar molecules across cell membranes leads to the multifaceted disease diabetes. SEE ALSO DIABETES; DISEASE, GENETICS OF; HEMOGLOBINOPATHIES; MUSCULAR DYSTROPHY; SIGNAL TRANSDUCTION.

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Polygenic Trait *See Complex Traits*

Polymerase Chain Reaction

The polymerase chain reaction (PCR) is a laboratory technique for “amplifying” a specific DNA sequence. PCR is extremely efficient and sensitive; it can make millions or billions of copies of any specific sequence of DNA, even when the sequence is in a complex mixture. Because of this power, researchers can use it to **amplify** sequences even if they only have a minute amount of DNA. A single hair root, or a microscopic blood stain left at a crime scene, for example, contains ample DNA for PCR.

PCR has revolutionized the field of molecular biology. It has enabled researchers to perform experiments easily that previously had been unthinkable. Before the mid-1980s, when PCR was developed, molecular biologists had to use laborious and time-consuming methods to identify, clone, and purify DNA sequences they wanted to study. Kary Mullis was awarded the 1993 Nobel Prize in Chemistry for inventing PCR.

PCR is based on the way cells replicate their DNA. During DNA replication, the two strands of each DNA molecule separate, and DNA polymerase, an enzyme, assembles nucleotides to form two new partner strands

amplify produce many copies of

for each of the original strands. The original strands serve as **templates** for the new strands. The new strands are assembled such that each nucleotide in the new strand is determined by the corresponding nucleotide in the template strand. The nucleotides adenine (A) and thymine (T) always lie opposite each other, as do cytosine (C) and guanine (G). Because of this base-pairing specificity, each newly synthesized partner strand has the same sequence as the original partner strand, and replication produces two identical copies of the original double-stranded DNA molecule.

In PCR, a DNA sequence that a researcher wants to amplify, called the “target” sequence, undergoes about thirty rounds of replication in a small reaction tube. During each replication cycle, the number of molecules of the target sequence doubles, because the products and templates of one round of replication all become the templates for the next round. After n rounds of replication, 2^n copies of the target sequence are theoretically produced. After thirty cycles, PCR can produce 2^{30} or more than ten billion copies of a single target DNA sequence. This is called a polymerase chain reaction because DNA polymerase catalyzes a chain reaction of replication.

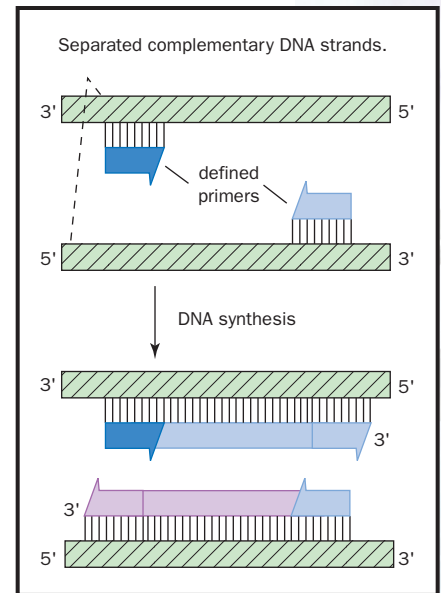
Designing Primers

To replicate DNA, DNA polymerases require not only a template, but also a primer. A primer is a sequence of single-stranded DNA that “anneals,” or binds, to the template by specific base-pairing. An automated apparatus called an oligonucleotide synthesizer, sometimes nickname a “gene machine,” can produce primers of any chosen sequence.

Primers for PCR are typically short sequences, around twenty nucleotides long. It is the primers’ sequences that are responsible for PCR’s enormous specificity. Researchers design primers so they are likely to bind to sequences on either side of the target DNA. They do so by making the primers complementary to the appropriate sequences and by making them long enough that they are unlikely to bind elsewhere.

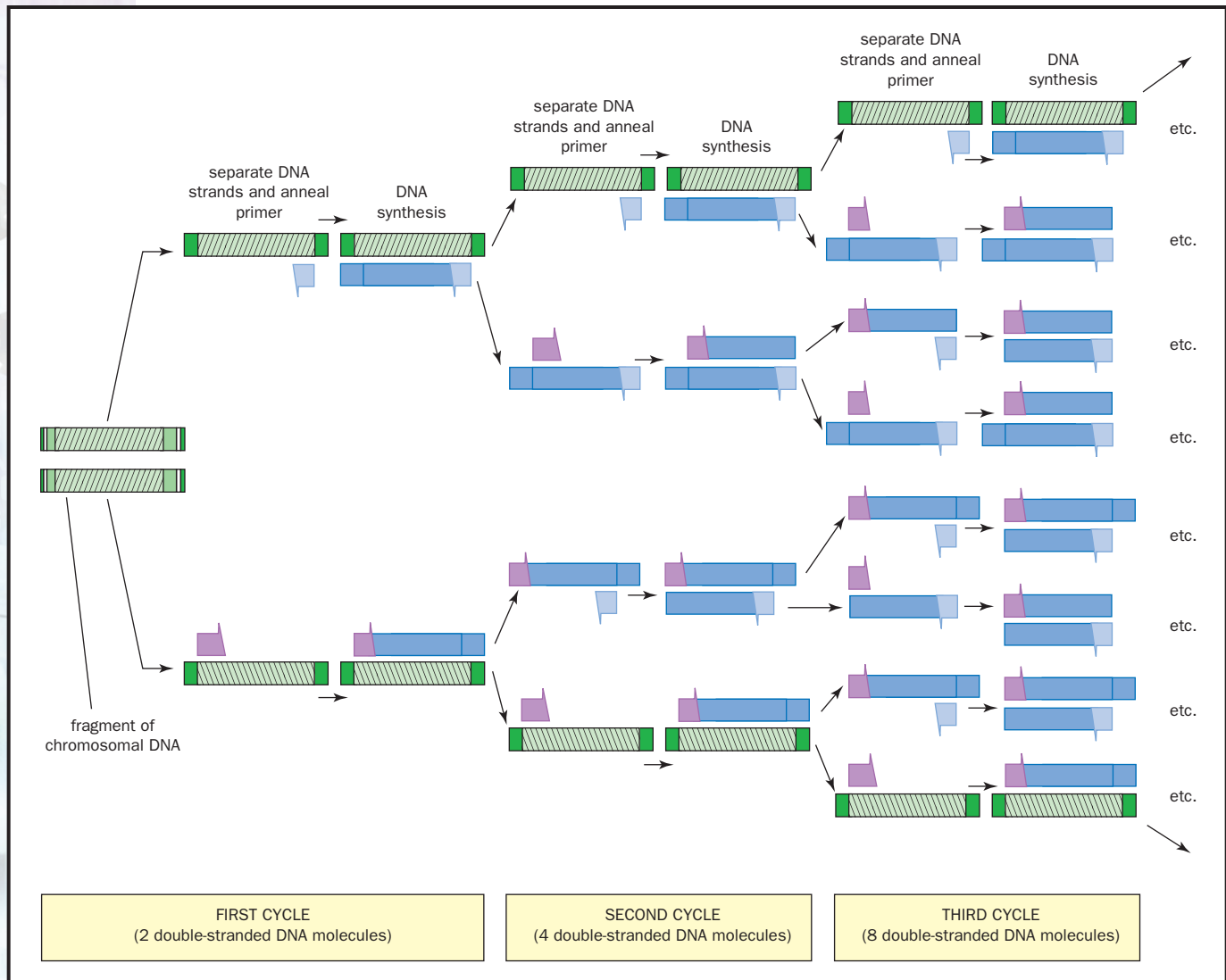
The longer the primer, the more likely it is that it will be complementary only to the target sequence. Because any single position in a DNA sequence can be occupied by either an A, T, C, or G, there is a one in four chance that any position will contain an A, for example. (This is an approximation, because the nucleotides are not distributed equally or randomly in DNA.) The odds that any specific DNA sequence that is n nucleotides long would be present at a given spot in a DNA sequence is therefore 1 in 4. The chance that a particular twenty-nucleotide sequence (a typical length for a PCR primer) would occur in a given spot at random is less than one in one trillion (10^{-12}). The human genome has only about three billion (3×10^9) nucleotide pairs, so any twenty-nucleotide-long sequence is very unlikely to occur more than once by chance in the human genome.

Researchers design two primers that will bind to opposite strands of the DNA on either side of the target sequence. They design them to “point” the right way, so that the section of DNA between, not outside of them, is copied. Designing the primers to “point” in the right direction simply requires building them so that their 3′ ends lie toward the target DNA and their 5′ ends lie away from it. The ends of any segment of DNA, including the complete strand, are chemically different.



PCR begins by separating complementary DNA strands, and ends by adding short primers that match opposite ends of complementary strands. Adapted from Alberts, 1995.

templates master copies



Multiple rounds of separation and copying can quickly produce billions of copies of the original DNA sequence. Adapted from Alberts, 1995.

microliters one thousandth of a milliliter

catalyze aid in the reaction of

buffers substances that counteract rapid or wide pH changes in a solution

One end is called the 5' (pronounced "5-prime") end. The other is called the 3' end. In DNA replication, nucleotides are always added to the 3' end of a growing strand of DNA. DNA synthesis is said to proceed in a 5' to 3' direction. The two complementary strands of DNA are anti-parallel, which means that they run in opposite directions. The 5' end of one strand lies next to the 3' end of the other, as shown in the diagram.

A Typical PCR Reaction

A typical PCR reaction consists of the following components, mixed together in a solution with a total volume of between 25 and 100 **microliters**. The solution must include the template DNA, the primers, nucleotides to serve as building blocks for the newly forming DNA, DNA polymerase to **catalyze** the synthesis, and **buffers** and salts, usually including magnesium, that are required for optimal activity of the DNA polymerase. The template can be an unpurified mixture of DNA, such as DNA extracted from a swab of cheek cells from a patient or crime suspect.

To perform the PCR reaction, the tube containing the solution is placed into a machine called a DNA thermal cycler. Thermal cyclers are basically programmable heating blocks. They usually contain a thick aluminum block with holes in which PCR reaction tubes can fit snugly. The block can be rapidly cooled or heated to specific temperatures, for specific lengths of time, under programmable computer control. Each cycle in a PCR reaction is controlled by changing the temperature of the block and, therefore, of the reaction mixture.

The first step in PCR is to heat the mixture to a high temperature, usually 94 to 95 °C, for about five minutes. The hydrogen bonds that hold together the two strands of a double helix are broken at these temperatures, and the DNA separates into single strands. This process is termed denaturation.

In the second step, the PCR mixture is cooled to a lower temperature, typically between about 50 °C and 65 °C. This allows the primers to **anneal** to their specific complementary sequences in the template DNA. The temperature for this step is chosen carefully to be just low enough to allow the primers to bind, but no cooler. A lower annealing temperature might allow the primers to bind to regions in the template DNA that are not perfect complements, which could lead to the amplification of non-specific sequences.

anneal join together

The optimal annealing temperature for a set of primers can be determined by a formula that is based on the nucleotide composition of the primers, but it is often a matter of trial and error to find the best annealing temperature. The annealing step usually takes about fifteen to thirty seconds, an amazingly short time considering that the primers must “scan” through the template DNA to find their proper binding sites.

In the third step, the reaction is heated again, usually to about 72 °C, the temperature at which the DNA polymerase is most active. Most enzymes are destroyed at 72 °C. In the early days of PCR, scientists used a DNA polymerase that was derived from the bacterium *Escherichia coli*, which itself is most active at human body temperature, 37 °C. But the *E. coli* polymerase was destroyed at the high temperatures required for the denaturation and annealing steps, and the polymerase therefore had to be added anew to the reaction, during each PCR cycle.

To solve this problem, scientists purified DNA polymerases from microorganisms that live in hot springs or in deep-sea thermal vents. These organisms’ enzymes are most active at high temperatures. The most commonly used enzyme for PCR is called Taq DNA polymerase, which was originally purified from the hot-spring bacterium *Thermus aquaticus*. (Most commercially available preparations today are recombinant versions, produced in engineered *E. coli* strains.)

At 72 °C, Taq DNA polymerase adds nucleotides to the 3’ ends of annealed primers at the rate of about two thousand nucleotides per minute. Therefore, to amplify a sequence that is one thousand nucleotides long, the primer extension step must last about thirty seconds at 72 °C. By the end of this step, each template strand has a new complementary strand. This completes the first cycle of the PCR reaction.

The cycle can be repeated, at that point, by restarting the denaturation step. In the next cycle, the original two DNA strands will serve again as



templates, as will the two newly synthesized strands. In this way, the number of templates has doubled, and it will double again with each successive cycle.

At the end of the reaction, the tube contains DNA fragments that are almost solely copies of the target DNA. The original template DNA mixture is still present, but for the purpose of most applications (with the exception of subsequent PCR experiments), it is present in negligible amounts compared to the PCR product. The amplified DNA can be analyzed by gel electrophoresis, **ligated** into a cloning vector, labeled for use as a hybridization probe, or used in numerous other experimental procedures.

ligated joined together

Contamination in PCR Reactions

The extreme sensitivity of PCR for amplifying rare DNA sequences is a mixed blessing. Just as PCR can easily amplify any sequence that a researcher wants to amplify, it can also amplify other sequences.

Amplifying a minute amount of DNA isolated from an ancient mosquito preserved in amber, for instance, could be extremely difficult. DNA from other sources could contaminate the sample during every step, including during the recovery of the amber, while researchers are drilling into it, and while the needle is prepared to remove the mosquito tissue. Contamination of the sample by even a single cell from another source can lead to amplification of that DNA, along with or instead of the mosquito's DNA. Especially if there are segments of contaminating DNA that are similar to the target DNA, primers may bind to the wrong segments.

Or consider a human geneticist who has designed a PCR assay to detect a particular genetic disease. Imagine that a positive result, amplification of the disease **allele** from a patient's DNA sample, would indicate the patient is a carrier for the disease. If even a trace of DNA from a disease carrier contaminates any of the PCR reagents, then assays performed on samples from non-disease carriers are likely to produce the diagnostic PCR amplification product. It isn't difficult to imagine that a lab that routinely performs this PCR assay might have lots of the tell-tale DNA contaminating benches, pipettes, and even lab coats.

allele a particular form of a gene

Two approaches address the contamination problem. First, laboratory practices for PCR aim to ensure the utmost cleanliness. Whole new industries have been created to produce contamination-resistant supplies, including micropipette tips. The second solution, as in all carefully planned experiments, is to use controls. Negative controls, including mixtures that have not had any template added or that contain a template known to lack the target sequence, are particularly important for PCR experiments.

PCR Applications and Variations

PCR is such a powerful, easy, and relatively inexpensive technique that it seems that molecular biologists are always looking for ways to use PCR in their research. Every month, scientific journals describe modifications to tailor the basic PCR approach to new applications.

One variation that has proved very fruitful in gene identification is the use of "degenerate primers." Many genes tend to be highly conserved among different species. Homologues, which are genes from different organisms

whose protein products have similar functions, tend to have very similar, but not necessarily identical, sequences. The differences in sequence make it challenging to design standard PCR primers to search for homologues.

However, by comparing the DNA sequences of the gene as it occurs in many different species and finding portions of the sequence that are the same in all the species, a researcher can make an educated guess regarding which nucleotides in an unidentified homologue are likely to be identical to those in a known homologue.

The researcher can design a set of “degenerate” PCR primers, which are primers whose nucleotide sequence is fixed only in those positions where the nucleotides are presumed to be known. In the other positions, nucleotides are allowed to incorporate at random. This makes it likely that at least one of the primers will amplify the unknown target. By conducting PCR with degenerate primer sets, and by using primer annealing temperatures that are lower than normal to allow for less-than-perfect base-pairing, a researcher can often amplify a gene in a single experiment, thus isolating the new homologue and allowing it to be sequenced and studied.

Another important variation on PCR is reverse-transcription PCR. This technique involves first copying RNA into DNA molecules, using the enzyme **reverse transcriptase**, and subsequently using the standard PCR technique to amplify this complementary DNA (cDNA). Because the messenger RNA content of a cell or tissue represents only the genes that are actively being expressed, this technique provides a powerful method of analyzing gene expression. SEE ALSO BIOTECHNOLOGY; CLONING GENES; DNA POLYMERASES; GEL ELECTROPHORESIS; GENETIC TESTING; HOMOLOGY; HUMAN GENOME PROJECT; NUCLEOTIDE; REPLICATION; SEQUENCING DNA.

reverse transcriptase
enzyme that copies RNA
into DNA

Paul J. Mublrud

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Polymorphisms

Genetic polymorphisms are different forms of a DNA sequence. “Poly” means many, and “morph” means form. Polymorphisms are a type of genetic diversity within a population’s gene pool. They can be used to map (locate) genes such as those causing a disease, and they can help match two samples of DNA to determine if they come from the same source. Depending on its exact nature, a polymorphism may or may not affect biological function.



A short tandem repeat (STR) polymorphism at a particular locus on a chromosome.



Coding and Noncoding Sequences

The amino acid sequence of proteins is directed by the information found in genes, which in turn are made up of DNA. Genes that have different DNA sequences are said to be polymorphic. These different gene forms are called alleles, exemplified by the alleles that control eye color. When alleles result in differences in the amino acid sequence of a protein, the proteins encoded by alleles are called isoforms. The position of the gene on a chromosome is its locus (plural, loci). More generally, a locus refers to any position on a chromosome, whether or not a gene is located there.

Polymorphisms arise through mutation. The mutation may be due to a change from one type of **nucleotide** to another, an insertion or deletion (collectively known as indels), or a rearrangement of nucleotides. Once formed, a polymorphism can be inherited like any other DNA sequence, allowing its inheritance to be tracked from parent to child.

Polymorphisms are also found outside of genes, in the vast quantity of DNA that does not code for protein. Indeed, regions of DNA that do not code for proteins tend to have more polymorphisms. This is because changes in DNA sequences that encode proteins may have a harmful effect on the individual that carries it. Polymorphisms that do not have any effect on the organism are said to be selectively neutral since they do not affect its ability to survive and reproduce.

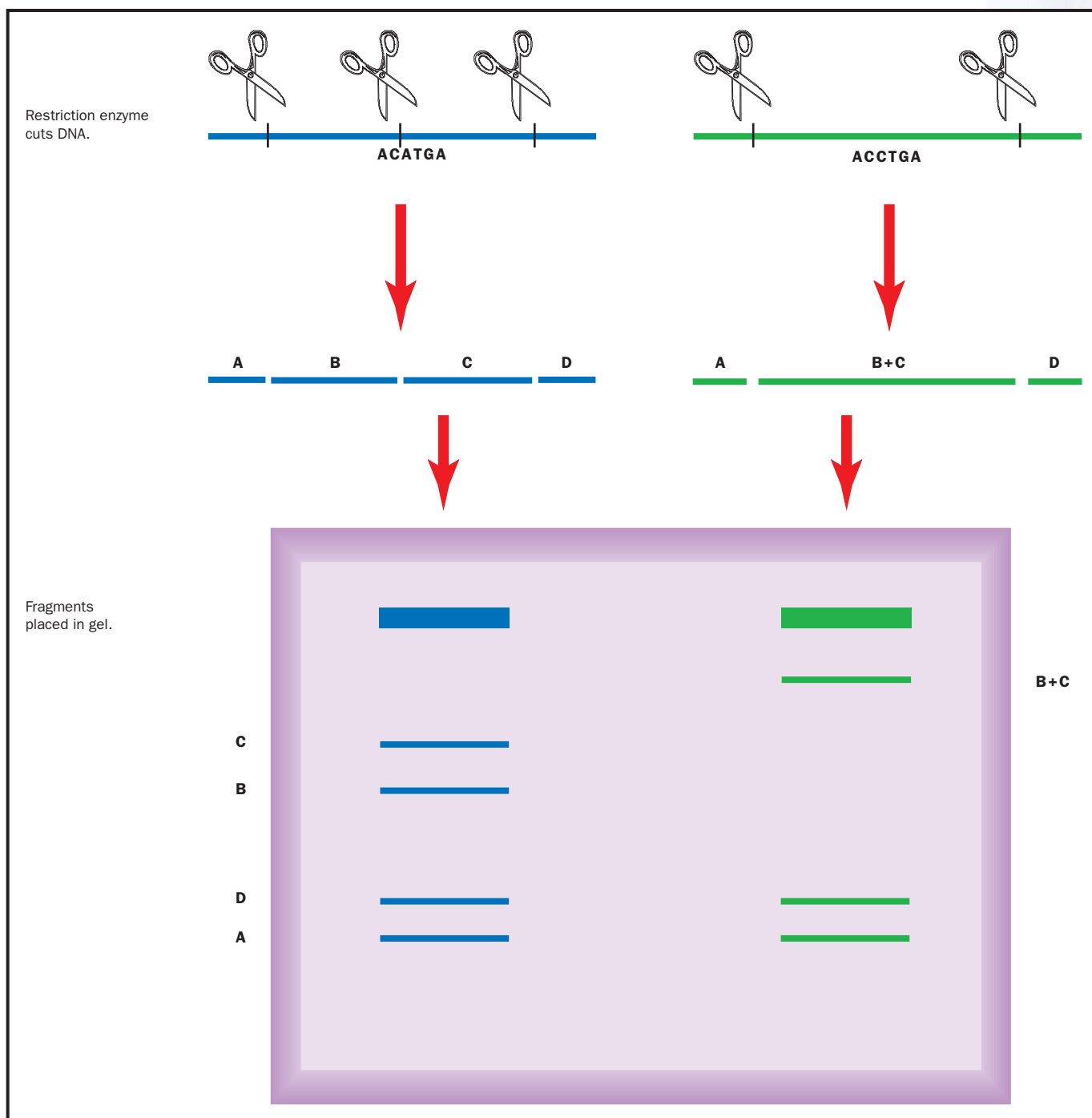
Identifying Polymorphisms

The process of determining an individual's genetic polymorphisms is known as genotyping. One of the earliest methods used in genotyping looked not at genes but at polymorphic proteins known as isoenzymes, or isozymes. Isoenzymes are different forms of a protein, with slightly different amino acid compositions. Since a protein's amino acid composition is genetically programmed by the DNA sequence that encodes it, analysis of isoenzymes surveys genetic polymorphism. Because these differences in amino acid composition can cause proteins to have different electrical charges, isoenzyme polymorphisms are assessed by extracting an organism's proteins and separating them using gel electrophoresis—a technique also used to study DNA polymorphisms.

In gel electrophoresis, an electric field is applied across a gel matrix, and molecules move through the matrix in response to the electric field. The gel matrix is a porous material, similar to Jell-O, that acts as a sieve and slows down large molecules more than small molecules. Isoenzymes move through the gel matrix according to their electrical charge and size, and are separated from each other on this basis. In this way, different isoenzymes can be identified.

Many tools for assessing DNA polymorphisms are now available. Some of these methods assess length polymorphism (indels), sequence polymorphism (base changes or rearrangements), or combinations of the two. DNA

nucleotide a building block of RNA or DNA



sequencing reveals all types of polymorphisms but is costly and labor-intensive. Gel electrophoresis is used in DNA analysis to both separate and sequence DNA. PCR (polymerase chain reaction) is frequently employed beforehand to produce large quantities of the DNA to be analyzed.

RFLPs

An early method of detecting DNA polymorphisms still in use employs restriction endonucleases. These bacterial enzymes cut DNA at specific recognition sequences. Restriction enzymes cleave DNA into a characteristic set

Restriction fragment length polymorphisms can be separated and identified by gel electrophoresis. A single nucleotide polymorphism (A→C) causes loss of a restriction site.



base pairs two nucleotides (either DNA or RNA) linked by weak bonds

of fragments that can be separated by gel electrophoresis. Some polymorphisms alter recognition sequences, so that the enzyme no longer recognizes a site or recognizes a new site. This results in a new set of DNA fragments that can be compared to others to detect the differences. These differences are called restriction fragment length polymorphisms (RFLPs).

STRs, VNTRs, and SNPs

Repetitive genetic elements are an important class of polymorphic DNA. These sequences consist of several repeats of a simple DNA sequence pattern, and they typically do not encode a protein or have strict requirements of size and sequence. For example, the two **base pairs** cytosine (C) and adenine (A) may be found together multiple times, resulting in a “CACACACA” sequence. If another copy of this sequence were found as “CACA” (two CA pairs shorter), then this sequence would be polymorphic. Repetitive genetic elements include microsatellites or STRs (short tandem repeats) and the minisatellites or VNTRs (variable number of tandem repeats), which are distinguished primarily on the basis of size and repeat pattern: The repeated sequence in microsatellites range from two to six bases, while in a VNTR it ranges from eleven to sixty base pairs.

Differences in single base pairs, known as single nucleotide polymorphisms (SNPs), are a valuable class of polymorphism that can be detected by DNA sequencing, RFLP analysis, and other methods such as allele-specific PCR and allele-specific DNA hybridization. Many RFLPs are due to single nucleotide polymorphisms. There are hundreds of thousands of SNP loci throughout the human genome, making them especially valuable for mapping human disease genes.

Uses of Polymorphisms

The study of polymorphism has many uses in medicine, biological research, and law enforcement. Genetic diseases may be caused by a specific polymorphism. Scientists can look for these polymorphisms to determine if a person will develop the disease, or risks passing it on to his or her children. Besides being useful in identifying people at risk for a genetically based disease, knowledge of polymorphisms that cause disease can provide valuable insight into how the disease develops. Polymorphisms located near a disease gene can be used to find the gene itself, through mapping. In this process, researchers look for polymorphisms that are co-inherited with the disease. By finding linked polymorphisms on smaller and smaller regions of the chromosome, the chromosome region implicated in the disease can be progressively narrowed, and the responsible gene ultimately can be located.

A related use of polymorphism is widely employed in agriculture. If a polymorphism can be identified that is associated with a desirable characteristic in an agriculturally important plant or animal, then this polymorphism can be used as a genetic flag to identify individuals that have the desirable characteristic. Using this technique, known as marker-assisted selection, breeding programs aimed at improving agriculturally important plants and animals can be made more efficient, since individuals that have the desired trait can be identified before the trait becomes apparent.

Polymorphisms can be used to illuminate fundamental biological patterns and processes. By studying polymorphisms in a group of wild animals, the familial relationships (brother, sister, mother, father, etc.) between them can be determined. Also, the amount of interbreeding between different groups of the same species (gene flow) can be estimated by studying the polymorphisms they contain. This information can be used to identify unique populations that may be important for survival of the species. Sometimes it is not immediately obvious if two different groups of organisms should be classified as different species. Comparing the genetic polymorphisms in the two groups aids in making a judgment as to whether they warrant classification as different species.

If enough polymorphisms are analyzed, it is possible to distinguish between individual humans with a high degree of confidence. This method is known as DNA profiling (or DNA fingerprinting) and provides an important tool in law enforcement. A person's genotype, or DNA profile, can be determined from very small samples, such as those that may be left at a crime scene (hair, blood, skin cells, etc.). The genotype of samples found at the crime scene can then be compared to a suspect's genotype. If they match, it is very likely that the suspect was present at the crime scene. Currently, the FBI uses thirteen different polymorphic loci for DNA fingerprinting. In a similar manner, analysis of polymorphisms can help prove or disprove fatherhood (paternity) in cases where responsibility for a child is disputed. SEE ALSO GEL ELECTROPHORESIS; LINKAGE AND RECOMBINATION; MAPPING; MUTATION; REPETITIVE DNA ELEMENTS.

R. John Nelson

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Polyploidy

In eukaryotic organisms, chromosomes come in sets. The **somatic** cells, called soma, usually have a **diploid** chromosome number, which in scientific notation is abbreviated as $2N$. The diploid state contains two sets of chromosomes, one set of which has been contributed by each parent. A single set of chromosomes composes the **haploid** chromosome number, which is abbreviated as N . The haploid set is found in reproductive cells or gametes (also called the germplasm). In humans the diploid number is 46, and is represented as $2N = 46$. Human sperm or eggs, however, have a haploid number of 23, which is represented as $N = 23$. In some circumstances, however, an organism can have more than two chromosomal sets. This occurrence is called polyploidy.

One cause of polyploidy is polyspermy. If two sperm fertilize an egg, the resulting zygote or fertilized egg will have three sets of chromosomes,

somatic nonreproductive; not an egg or sperm

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

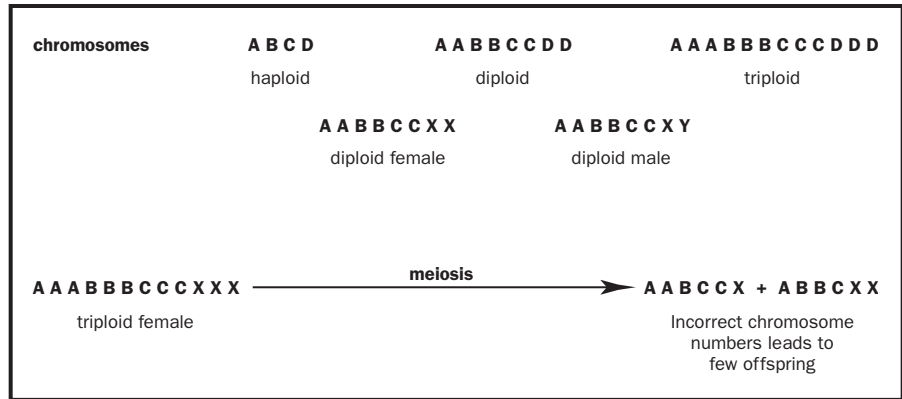
haploid possessing only one copy of each chromosome



Ploidy refers to the number of chromosomes in an organism's genome. A haploid organism has just one copy of each chromosome (A, B, C, and D). Diploids have two of each, and may have sex chromosomes (X and Y). Triploids have three, which prevents proper pairing and separation of chromosomes in meiosis.

triploid possessing three sets of chromosomes

autosomes chromosomes that are not sex determining (not X or Y)



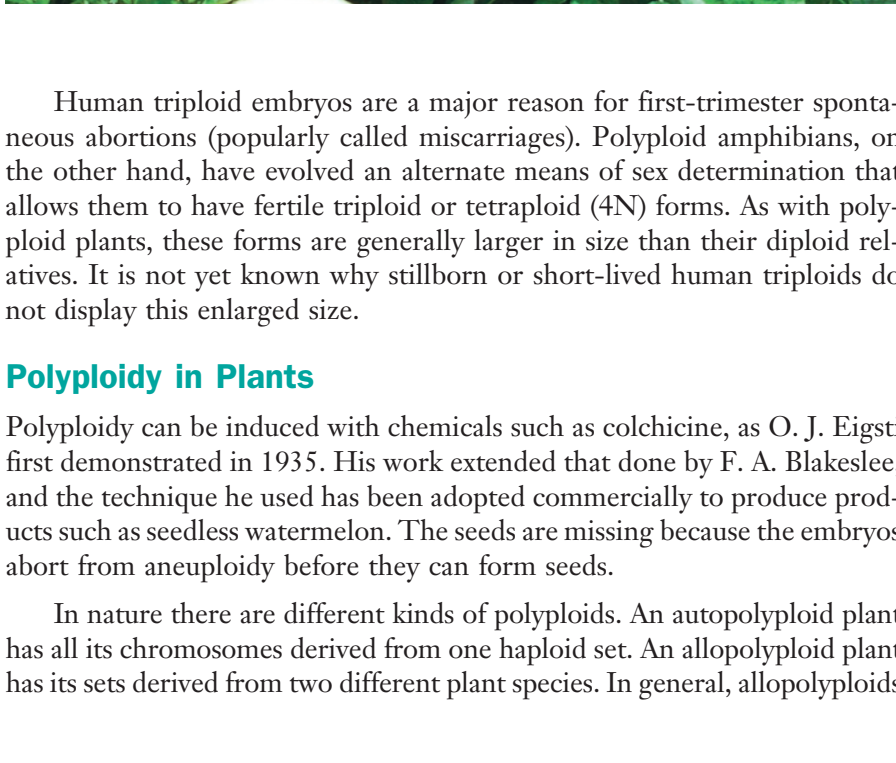
and thus have a **triploid** number ($3N$). When this occurs in humans, $3N = 96$. Triploidy in humans and most other animals is incompatible with life. Triploid individuals abort or fail to survive the first days of life after birth. Polyploidy is more common in plants, and polyploid forms often survive to produce much larger cells and plant organs. Ferns, which may have up to 1,500 chromosomes, are frequently polyploid, as are varieties of domesticated cereal plants. Most often, polyploids run in sets of three to eight (triploid to octoploid).

Polyploidy in Animals

Geneticist Hermann Muller argued that polyploidy is more rare in animals than plants because animals have a more complex development, with more organ systems that are fine-tuned to dosages of genes. Any given gene is represented three times in a triploid. If the amount (dosage) of gene product causes a heart, brain, or other vital organ not to form, the embryo will abort. When these developmental genes produce too much or too little of the products that induce organ formation, as they might if there are too many or too few copies of the genes, events occur too soon or too late to be coordinated. Muller raised the possibility that the sex chromosomes serve as a barrier to polyploidy in most animals. Plants, by contrast, do not usually have sex chromosomes, and thus this sexual reproductive barrier is not a problem for them.

Muller noted that most animals use a sex-chromosome mechanism for sex determination. In fruit flies and humans, diploid males have the sex chromosomes XY, whereas diploid females have XX. A triploid fly or human would have three chromosomes along with three sets of **autosomes**. In such a triploid, XXX will result in a female. However, a zygote having XXY XYY may not produce a male. Rather, it may result in an intersex organism, with abnormal mixed male and female reproductive organs.

While human triploids do not survive, this is not the case for fruit flies. The XXY or XYY is an intersex, sterile form, but the triploid female is fertile. If the $3N$ female is mated to a $2N$ XY male, however, only a relatively few offspring will emerge, because many of the eggs will have an incorrect number of chromosomes. This state of excesses or deficits of chromosomes in an otherwise diploid or triploid cell is called aneuploidy. Aneuploid embryos rarely survive in humans or other animals, although there are exceptions (such as infants born with Down syndrome).



Polyploidy in Plants

Polyploidy can be induced with chemicals such as colchicine, as O. J. Eigsti first demonstrated in 1935. His work extended that done by F. A. Blakeslee, and the technique he used has been adopted commercially to produce products such as seedless watermelon. The seeds are missing because the embryos abort from aneuploidy before they can form seeds.

In nature there are different kinds of polyploids. An autopolyploid plant has all its chromosomes derived from one haploid set. An allopolyploid plant has its sets derived from two different plant species. In general, allopolyploids

Polyploidy can be induced

In nature there are different kinds of polyploids. An autopolyploid plant has all its chromosomes derived from one haploid set. An allopolyploid plant has its sets derived from two different plant species. In general, allopolyploids

gamete reproductive cell, such as sperm or egg

are fertile and survive, whereas autopolyploids are sterile and must be propagated as clones (identical twins), by cuttings.

The difference between autopolyploidy and allopolyploidy can be appreciated by an example. No one knows the reasons for mitotic failure leading to spontaneous tetraploids, but artificial ones are induced by mitotic poisons, like colchicine, that prevent spindle fiber formation. If one species has chromosomes ABCD in a (haploid) **gamete**, and a related species has chromosomes FGHI, the resulting (diploid) zygote will have a chromosome set consisting of ABCDFGHI. If that collection of chromosomes undergoes a spontaneous doubling, the resulting plant is AABBCCDDFFGGHHII. Such a plant will produce ABCDFGHI gametes and by self-pollination, which is common in many flowering plants, the new allopolyploid will be fertile.

In the case of autopolyploids, by contrast, the chromosomes ABCD become triplicated (3N: AAABBBCCCDDD) or quadruplicated (4N: AAAABBBBCCCCDDDD). This may lead to nondisjunctional separations during meiosis, wherein the chromosomes will divide improperly or incompletely. In the 3N plant many of the gametes may be AABCDD or ABCCD or other variations of aneuploidy that will disturb embryonic development.

Among familiar plant polyploids are strains of wheat with chromosome numbers of 14 (2N), 28 (4N), and 42(6N), all of which are based on an ancestral form whose haploid number was 7. Chrysanthemums have a series of varieties with a range of chromosome numbers: 18, 36, 54, 72, and 90. The ancestral haploid is assumed to be 9. About half of all flowering plant species are believed to have polyploid varieties. If an accidental doubling of the zygote chromosome number is the major mechanism involved, most of these forms are tetraploid.

Genetic Analysis

The transmission of genetic traits in polyploids is more difficult to calculate than in diploids because a gene for a recessive trait in a triploid, for example, would have to appear in the same location on all three of its homologous chromosomes in order for it to be phenotypically apparent. Such calculations, when done for diploids, rely upon binomial equations and generate a familiar ratio of 9AB:3aB:3Ab:1ab, whereas the calculations for polyploid plants require the use of trinomial equations for triploids and quadrinomial equations for tetraploids, instead of the traditional binomial $(A + B)^2$ that generates the familiar 9AB:3aB:3Ab:1ab ratio for diploids. Thus for a trinomial (three gene) the equation will be the expansion of $(A + B + C)^3$.

The use of polyploids in laboratory research has allowed research into the function of specific genes. For instance, triploid female fruit flies crossed to diploid males were used to create a diploid offspring with a chromosome of a sibling species. In this experiment, the tiny fourth chromosome of *Drosophila simulans* was inserted into an otherwise diploid *D. melanogaster* offspring. This permitted analysis of the genes shared in common (most of them) as well as gene differences that led to visible malformations in the hybrid fly. Triploid flies have also been crossed to irradiated diploid males to prove that X rays induce breaks in chromosomes, causing **apoptosis** and

apoptosis programmed cell death

embryonic abortion. SEE ALSO CHROMOSOMAL ABERRATIONS; MEIOSIS; MULLER, HERMANN; SEX DETERMINATION; X CHROMOSOME, Y CHROMOSOME.

Elof Carlson

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Population Bottleneck

A population bottleneck is a significant reduction in the size of a population that causes the extinction of many genetic lineages within that population, thus decreasing genetic diversity. Population bottlenecks have occurred in the evolutionary history of many species, including humans. Present-day bottlenecks are seen in endangered species such as the Yangtze River dolphin, whose numbers have dwindled to less than 100. Endangered species that do not become extinct may expand their numbers later on, but with a limited amount of genetic diversity with which to adapt to changing conditions. The genomes of future populations will reflect the narrowing of genetic possibility for thousands of years.

Reconstructing Genealogies

The genomes of living organisms record both genealogical and population histories. Our own genome tells a remarkable story of events in recent human evolution. Relatedness of individuals within and between populations and species can be determined by measuring the number of genetic differences between two individuals. When applied to segments of the genome that accumulate mutations at relatively constant rates over time, they can provide information about the time that has elapsed since the existence of their last common ancestor. Research shows that human and chimpanzee lineages diverged about six million years ago, that neanderthals and anatomically modern humans diverged 500 thousand years ago, and that all living humans can trace their ancestry to a maternal lineage that lived in Africa about 130 thousand years ago. Figure 2 illustrates differentiation of lineages and the effects of bottlenecks on diversity.

Reconstructing Ancient Population Sizes

Knowledge of mutation rates also permits reconstruction of past population sizes. A small number of genetic differences between individuals in a population or species may indicate either a recent origin, or a population bottleneck. Which of these two possible causes is responsible can be determined by measuring the number of so-called pairwise differences (mismatch distributions) in the DNA sequences that occur between individuals. Population expansion times are earlier for populations with higher average pairwise differences. Irregular mismatch distributions indicate long-term populations that have been stable for long times.

As shown in Figure 3, humans have remarkably little genetic diversity, especially in comparison to our closest living relative, the chimpanzee.



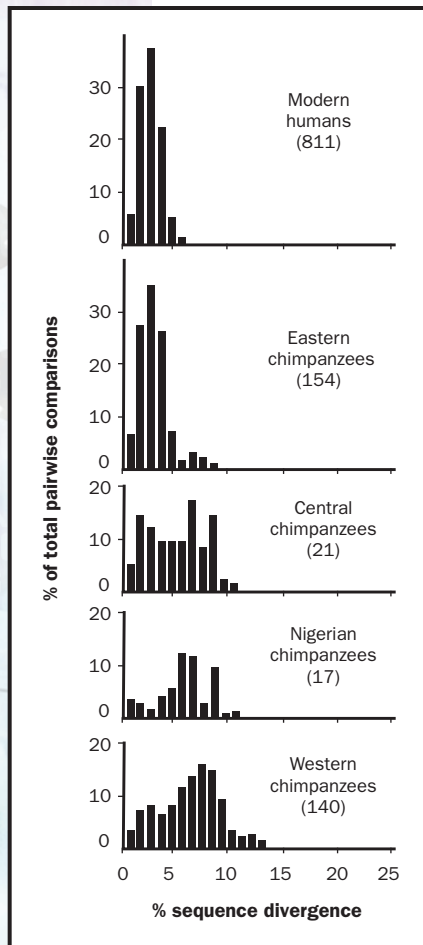


Figure 1. Pairwise sequence divergence distributions in humans and four geographic populations (subspecies) of chimpanzees. The broad and uneven mismatch distributions of central, Nigerian and Western chimpanzees indicate they have maintained relatively large effective population sizes for a very long time. The low genetic diversities of eastern chimpanzees and humans reflect recent population expansions following the loss of many lineages during population bottlenecks. The small peak on the right side of the Eastern chimpanzee distribution corresponds to a population expansion 69 thousand years ago, and the larger peak reflects expansion around 20,000 years ago, near the end of the last ice age.

Indeed, there is substantially more genetic difference among individuals within chimpanzee troops in West Africa than among all living humans on earth. As shown in Figure 1, this is due to a series of bottlenecks in human evolutionary history. Geneticists studying many different parts of the human genome have concluded that the past effective population size (that is, the number of reproducing females) averaged only 10,000 individuals over the last one million years, and was as low as 5,000 around 70,000 years ago. Compare this to the approximately one billion reproducing females alive today, and it becomes clear just how narrow these bottlenecks were.

Population Bottlenecks and Expansions in Human Evolution

The genetic structure of human populations suggests four bottlenecks in our lineage. Stanley Ambrose has proposed that two bottlenecks may be related to past environmental changes. Marta Lahr has attributed bottlenecks to migrations of small populations across geographic barriers, a phenomenon variously referred to as the founder effect or colonization bottlenecks.

Bottleneck 1. When traced backward in time, all human lineages coalesce to an ancestral lineage that lived in Africa about 130 thousand years ago. This date coincides with the end of the penultimate glacial period (190 to 130 thousand years ago). Populations were probably very small during this ice age. Expansion (bottleneck release) occurred during the last interglacial (130 to 71 thousand years ago), when warm climates and higher rainfall returned. Other lineages probably existed at that time, but they left no modern descendants.

Bottleneck 2. A severe bottleneck around 70,000 years ago may have reduced the effective population size in Africa to only 5,000 females. This date coincides with the super-eruption of Toba, a volcano located in northern Sumatra. Toba blasted over 800 cubic kilometers of volcanic ash and millions of tons of sulfur gas into the atmosphere. The volcanic ash settled relatively quickly, but the sulfur formed a long-lasting stratospheric haze that reflected sunlight and may have caused rapid global cooling. Annual layers of ice in the Greenland ice sheet suggest that this haze lasted six years, causing a “volcanic winter.” This was followed by 1,000 years of the coldest temperatures of the last ice age. Analysis of air trapped in these ice layers suggests that temperatures dropped 16 °C over Greenland during this “instant ice age.” Drought and famine during this cataclysmic event undoubtedly decimated populations in most parts of Africa.

Bottleneck 3. Analysis of Y chromosomes shows that all modern populations in southern Australasia can trace their ancestry to a small founding population from the Horn of northeast Africa (Ethiopia and Somalia) around 60,000 to 70,000 years ago. Increases in windblown dust in Greenland ice indicate a rapid drop in sea level to more than 100 meters lower than at present. This would have greatly facilitated dispersal from Africa to the Arabian Peninsula. Expansion around the perimeter of the Indian Ocean culminated in the colonization of Australia about 60,000 years ago.

Bottleneck 4. Analyses of gene sequences provide evidence of a possible second exodus from Africa by a small founding population that traveled over-

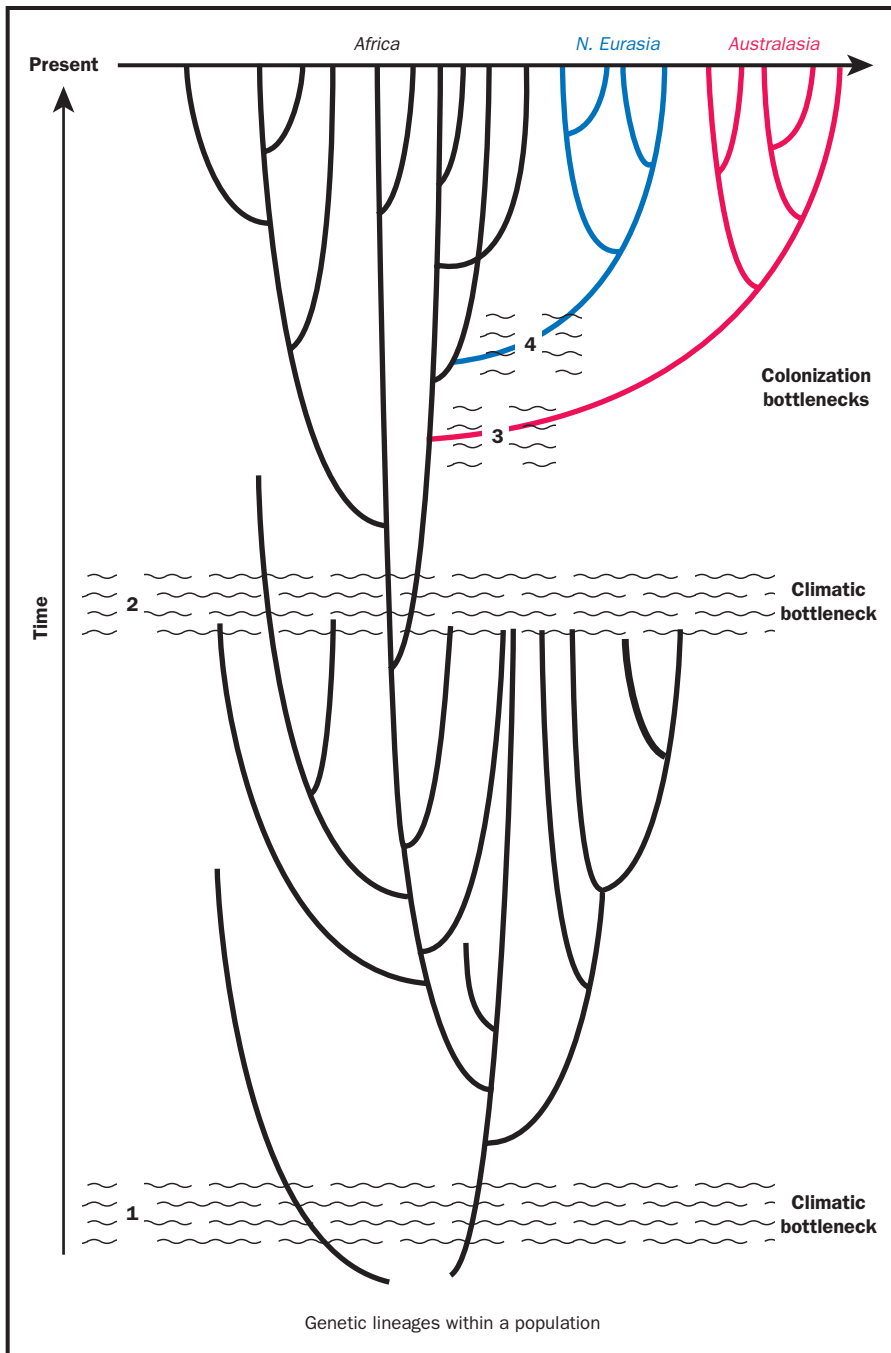


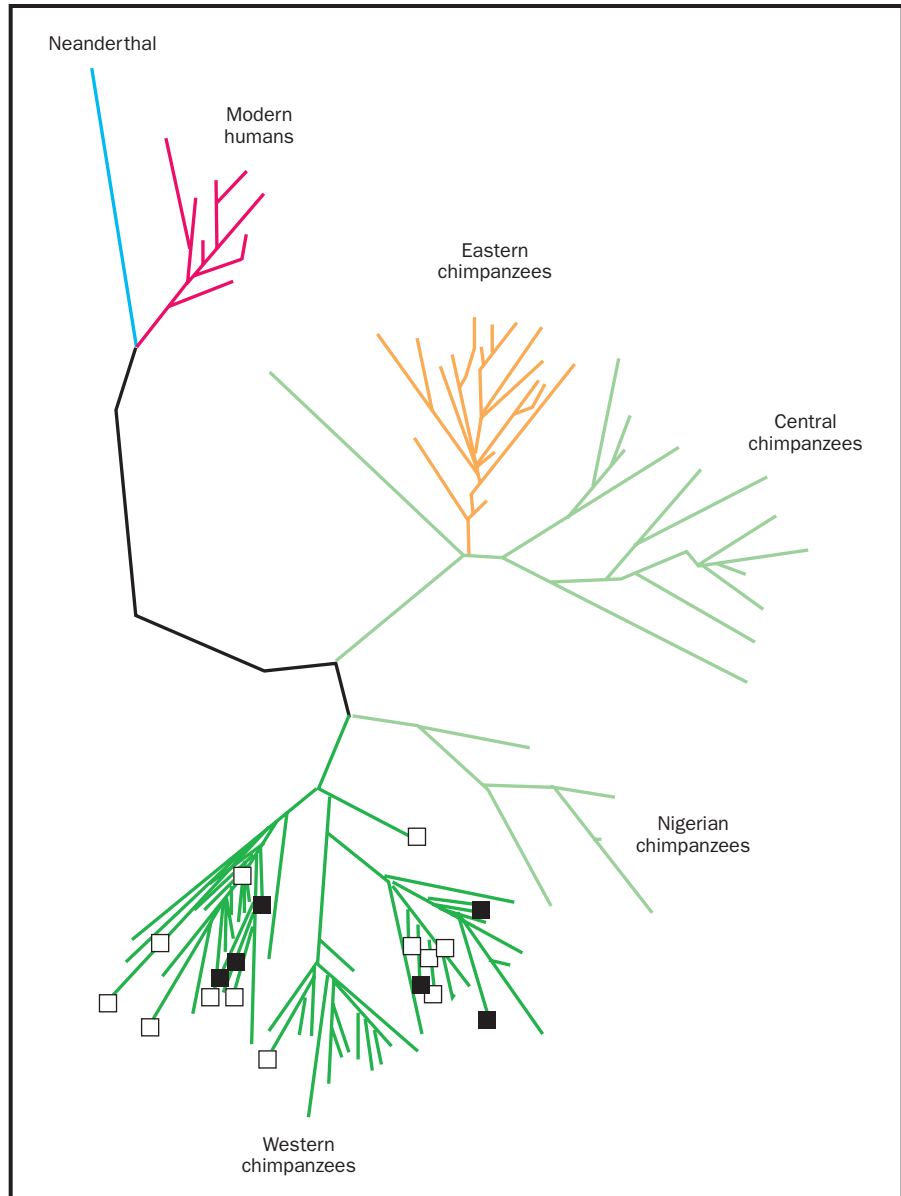
Figure 2. Differentiation of human genetic lineages within a population through time, and the effects of environmental and colonization bottlenecks (founder events) on survival of human lineages.

land via the shoreline of the Red Sea. This colonization bottleneck occurred during a period of milder climate about 50,000 years ago, and also coincides with the appearance of advanced stone tool technologies. Expansion continued into Europe and northern Asia. All living humans outside of Africa can thus trace their ancestry to these colonizing populations.

Technological and Social Influences on Past Population Size

Social and technological innovations in Africa during the later Middle Stone Age and early Later Stone Age (50,000 to 70,000 years ago) may

Figure 3. Unrooted family tree diagram of mitochondrial DNA control region sequences of a worldwide sample of 811 humans, and 332 chimpanzees from four regions of Africa. Branch length is proportional to the amount of genetic difference between individuals, and to the time since they shared a common ancestor. Symbols indicate individual chimpanzees belonging to the same social group. West African chimpanzees have three common ancestral lineages. Genetic differences between individuals in the same group substantially exceed differences between all living humans, and between humans and Neanderthals.



have facilitated population expansions and colonizations by enhancing survival in arid, unpredictable ice age environments. New stone tool technologies may have increased foraging efficiency and food supply. A system of mutual reliance and cooperation between distant foraging groups, mediated by reciprocal gift exchange, may have also increased humans' ability to survive in unpredictable environments. Further social and technological innovations may have facilitated population expansion within Africa, dispersals out of Africa, and the replacement of archaic populations, including Neanderthals, by anatomically modern humans outside of Africa.

Low levels of modern human diversity thus reflect our recent African ancestry and the effects of several population bottlenecks. In a similar fashion, colonization bottlenecks promoted rapid differentiation of northwestern Eurasians and southeastern Australasians. SEE ALSO CONSERVATION BIOLOGY:

GENETIC APPROACHES; FOUNDER EFFECT; HARDY-WEINBERG EQUILIBRIUM;
MOLECULAR ANTHROPOLOGY; POPULATION GENETICS; Y CHROMOSOME.

Stanley Ambrose

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Population Genetics

Population genetics is the study of the genetic structure of populations, the frequencies of **alleles** and **genotypes**. A population is a local group of organisms of the same species that normally interbreed. Defining the limits of a population can be somewhat arbitrary if neighboring populations regularly interbreed. All the humans in a small town in the rural United States could be defined as a population, but what about the humans in a suburb of Los Angeles? They can interbreed directly with nearby populations, and, indirectly, with populations extending continuously north and south for a hundred or more miles. In addition, a large human population often consists of subpopulations that do not readily interbreed because of differences in education, income, and ethnicity. Despite these complexities, one can make some simple definitions.

Gene Pool and Genetic Structure

All of the alleles shared by all of the individuals in a population make up the population's gene pool. In diploid organisms such as humans, every gene is represented by two alleles. The pair of alleles may differ from one another, in which case it is said that the individual is "heterozygous" for that gene. If the two alleles are identical, it is said that the individual is "homozygous" for that gene. If every member of a population is homozygous for the same allele, the allele is said to be fixed. Most human genes are fixed and help define humans as a species.

The most interesting genes to geneticists are those represented by more than one allele. Population genetics looks at how common an allele is in the whole population and how it is distributed. Imagine, for example, an allele "*b*" that when homozygous, "*bb*," produces blue-eyed individuals. Allele *b*

alleles particular forms of genes

genotypes sets of genes present

A caribou herd is a population: a group of organisms of the same species that inhabit an area and interbreed. Individuals may migrate between populations, causing gene flow.



might have an overall frequency in the population of 20 percent; that is, 20 percent of all the eye-color alleles are *b*.

However, not everyone who has the *b* allele will be homozygous for *b*. Some people will have *b* combined with another allele, “*B*,” which gives them brown eyes (because *B* is dominant and *b* is recessive). Others won’t have the *b* allele at all and instead will be homozygous for *B*.

The frequency of each genotype—whether *bb*, *Bb*, or *BB*—in the population is also of interest to population geneticists. The frequency of alleles and genotypes is called a population’s genetic structure. Populations vary in their genetic structure. For example, the same allele may have a frequency of 3 percent among Europeans, 10 percent among Asians, and 94 percent among Africans. Blood types vary across different ethnic groups in this way. The frequency of genotypes depends partly on the overall allele frequencies, but also on other factors.

Hardy-Weinberg Theorem

Large, isolated populations whose members mate randomly and do not experience any selection pressure will tend to maintain a frequency of genotypes predicted by a simple equation called the Hardy-Weinberg Theorem. For example, if *b* has a frequency of 20 percent and *B* has a frequency of 80 percent, we can predict the frequency of the three genotypes (*bb*, *Bb*, and *BB*). The total of all the genotype frequencies is 100 percent (*b* + *B*), and the frequencies of each are given by $(b + B)^2 = 100$ percent. This can be restated as the following equation:

$$100\% = b^2 + 2(bB + B^2).$$

And we can calculate the genotype frequencies as:

$$100\% = (20\%)^2 + 2(20\% \times 80\%) + (80\%)^2 = 4\% + 32\% + 64\%.$$

So even though 20 percent of all the genes in this imaginary population are *b* alleles, only 4 percent of the population is homozygous for *b* and actually



has blue eyes. Furthermore, this same distribution will be maintained over time, as long as the conditions of the Hardy-Weinberg Theorem are met.

However, few, if any, natural populations (including human ones) actually conform to the assumptions of Hardy-Weinberg, so both genotype frequencies and allele frequencies can and do change from generation to generation. For example, humans do not mate randomly. Instead, they tend to take partners of similar height and intelligence. And even in modern human populations, genetic diseases such as Tay-Sachs kill children long before they grow up and reproduce. A difference in survival and reproduction due to differences in genotype is called selection. Even subtle selection can change gene frequencies over long periods of time.

Another assumption of the Hardy-Weinberg theorem is that individuals from different populations do not mate, so that gene flow, the passage of new genetic information from one gene pool into another, is zero. Such isolation does characterize many animal and plant populations, but almost no modern human populations are isolated from all other populations. Instead, humans travel to different countries, inter-

Trains and buses ferry more than seven million commuters to and from Bombay every day. Demographers predict that the population of Bombay and its surrounding regions will exceed twenty-seven million by 2015.